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**DESIGN AND OPTIMIZATION OF AN ISOSORBIDE PRODUCTION
REACTOR FROM SORBITOL
FINAL REPORT**

GROUP NO: 6

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ABSTRACT

In this study, a continuous liquid-phase catalytic Plug Flow Reactor (PFR) with an annual production capacity of 5,000 tonnes has been designed for the production of isosorbide from sorbitol. Based on literature findings, H-Beta zeolite was chosen as the catalyst, and the process was modelled using a first-order kinetic model comprising serial and parallel reactions. It was assumed that the catalyst effect was included in the kinetic rate constants and was not modelled separately. The reaction system was analysed at 393.15 K, 403.15 K and 423.15 K. The results showed that an increase in temperature increased the reaction rate and significantly reduced the residence time required. Under the conditions examined, the optimum operating temperature was determined to be 423.15 K; at this temperature, the sorbitol conversion, isosorbide yield and selectivity were calculated to be 99.98%, 74.92% and 3.448%, respectively. The corresponding residence time was found to be 0.033 hours. Reactor sizing under optimal conditions showed that the required reactor volume was 0.1903 m³. The results obtained demonstrate that the continuous liquid-phase PFR using H-Beta zeolite can achieve high conversion and yield, while maintaining a small reactor size; this makes it a suitable design alternative for isosorbide production on an industrial scale.

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NOMENCLATURE

V Flow rate of liquid (**kmol/s**)

1. INTRODUCTION

The gradual depletion of fossil resources and the growing need for sustainable production processes have made the development of chemicals derived from biomass a priority. Isosorbide, one of these bio-based platform chemicals, is a renewable compound obtained through the double dehydration of sorbitol, which is produced from glucose derived from biomass, and has attracted considerable interest in recent years [4,5]. In addition, due to its low toxicity, high thermal stability and robust bicyclic structure, it is regarded as an important platform chemical in research into sustainable process development [6]. Isosorbide is a bio-based monomer used in the production of high-performance polymers; it also has various applications in the manufacture of pharmaceuticals, surfactants and plasticisers [6,7].

The conversion of sorbitol to isosorbide takes place via a complex reaction network comprising a series of sequential reaction steps [5]. In the reaction mechanism, sorbitol is first converted into sorbitan derivatives, which act as intermediates, and isosorbide is then formed by a second dehydration step [9]. However, various side reactions also occur within the reaction system, which leads to the formation of undesirable products [3]. Consequently, a high sorbitol conversion rate does not always equate to a high isosorbide yield; therefore, parameters such as yield and selectivity must be considered together when evaluating process performance [3].

Research in the literature indicates that continuous liquid-phase reactor systems offer significant advantages for the industrial-scale production of isosorbide [1]. In particular, continuous-flow Plug Flow Reactor (PFR) systems provide higher production efficiency, lower reactor volume requirements and easier scalability compared to traditional batch processes [1,2]. Also, it has been observed that H-Beta zeolite catalysts exhibit high isosorbide selectivity due to their suitable pore structures and acidic properties [10]. On the other hand, it has been noted that mesoporous carbon-based acid catalysts are among the alternative catalysts that provide high conversion and high product yield [13].

The aim of this study is to design and optimise a continuous liquid-phase catalytic PFR system with an annual production capacity of 5,000 tonnes of isosorbide. Within the scope of the project, temperature was selected as the design parameter, and its effects on sorbitol conversion, isosorbide yield, selectivity, residence time and reactor volume were investigated. Using the results obtained, optimal operating conditions were determined, taking into account both technical performance and process economics, and a final reactor design was proposed.

2. DESIGN SPECIFICS

2.1. Selection of Reactor Type

2.1.1. Reactor Selection

In this study, a Plug Flow Reactor (PFR) was selected as the reactor type for the conversion of sorbitol to isosorbide. The literature indicates that the sorbitol dehydration reaction involves a complex network of both sequential and parallel reactions. For this reason, reactor systems that can provide high product selectivity are preferred.

PFR has been evaluated as a suitable option for this process due to its suitability for continuous production, high conversion efficiency, and ability to increase the selectivity of the desired product in sequential reactions. Additionally, PFR systems can be easily scaled up to an industrial scale and integrated into continuous production lines. The literature review conducted in the interim report also concluded that continuous liquid-phase PFR systems are the most suitable reactor type for the production of isosorbide from sorbitol.

Reactor is considered as catalytic liquid-phase PFR packed with a solid acid catalyst. Based on interim literature survey, MC-PhSO₃H or H-Beta zeolite mesoporous carbon-based acid catalyst may be considered suitable catalyst options. And these catalysts provide Brønsted acid sites needed for sorbitol dehydration and may improve selectivity by reducing effect of the water accumulation and suppressing the side reactions. In kinetic calculations of this final design, the catalyst effects were assumed to be included in given apparent first order rate constants. So therefore, catalyst loading was not optimized as independent parameter, the assigned design parameter for Group 6 was reaction temperature.

2.1.2. Operating Conditions

In this study, it is assumed that the reactor operates in the liquid phase and in a steady state. It is assumed that the flow exhibits ideal plug flow behaviour, with complete mixing in the radial direction and negligible mixing in the axial direction. The design target was the production of 5,000 tons/year of high-purity isosorbide. Yearly operating time of 330 days/year and 24 hours/day was used, corresponding to the 7920 h/year. The initial sorbitol concentration was taken as 1 mol/L, it is consistent with MATLAB and Excel calculations. To investigate temperature-dependent changes in reactor performance, analyses were performed at temperatures of 393.15 K, 403.15 K, and 423.15 K.

Additionally, it was assumed that the fluid density remained constant across the reactor and that the system operated under isothermal conditions. The effects of pressure drop and internal and external mass transfer resistances on reactor performance were neglected.

The operating pressure was not directly included in kinetic model. However, for final design, pressure should be high enough to maintain liquid-phase operation and stable flow through catalytic bed. Pressure drop was neglected in preliminary calculation. In detailed industrial design, pressure drop across the packed catalyst bed must be checked using appropriate packed-bed pressure drop relation such as Ergun equation.

2.1.3. Assumptions

The following assumptions were made in the development of the reactor model:

- The reactor is operating in a steady-state.
- The flow exhibits ideal plug flow behaviour. (Mixing in the radial direction is complete, and mixing in the axial direction is negligible.)
- All reactions are irreversible.
- All reactions are first-order reactions with respect to the concentration of the reactants.
- The fluid density is constant.
- There is no change in the physical properties along the length of the reactor.
- The pressure drop has been neglected.
- The external mass transfer resistance has been neglected.
- Internal diffusion resistance has been neglected.
- The reactor is operating isothermally.
- Side reactions are limited to the reaction network specified in the project guidelines.
- Catalyst deactivation has been neglected.
- Catalyst activity remains constant throughout the reactor's operation. It is assumed that the reactor operates in the presence of a solid acid catalyst, and the catalytic effect is expressed using the kinetic constants specified in the project data.
- The selected catalyst is a solid acid catalyst, preferably H-Beta zeolite or MC-PhSO₃H mesoporous carbon-based catalyst.
- It has been assumed that the reactor operates for 330 days a year, taking into account the time required for filling, emptying and maintenance.
- Feed stream was assumed to contain only sorbitol; therefore, the initial concentrations of 1,4-sorbitan, isosorbide and undesired products were taken as zero.

- Sorbitol conversion and isosorbide yield were calculated based on initial sorbitol concentration $C_{S0} = 1 \text{ mol/L}$.
- Total undesired product concentration is calculated as a sum of the side product concentrations included in given reaction network $C_{S1,5} + C_{O1} + C_{O2}$.
- Since isosorbide yield is lower than 100%, required inlet sorbitol flow rate was calculated by dividing target isosorbide molar production rate by the calculated isosorbide yield.
- For preliminary reactor dimensioning reactor length to diameter ratio is assumed as $L/D = 10$.

2.2. Preliminary Design and Methodology

2.2.1. Design Basis and Design Objectives

The aim of this study is to design a suitable reactor system for the annual production of 5,000 tonnes of isosorbide from sorbitol. Following a literature review, the Plug Flow Reactor (PFR) was selected due to its suitability for continuous production and its ability to provide high selectivity. The design parameter specified for this project is the reaction temperature, and the effect of temperature on reactor performance and isosorbide yield has been investigated.

The primary objective of the optimisation study is to maximise isosorbide yield while keeping the reactor volume as small as possible. Reaction temperature has been selected as one of the most important operating variables, as it directly affects both reaction rates and product distribution. Since an increase in temperature can increase reaction rates while also potentially affecting the formation of side reactions, determining the optimum temperature is of great importance.

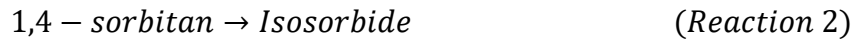
Analyses were conducted at three different temperatures as 393.15 K, 403.15 K and 423.15 K in accordance with the kinetic data provided in the project guidelines. For each temperature, sorbitol conversion, isosorbide yield and the formation of undesirable products were evaluated; furthermore, the required residence times and reactor volumes were calculated. Finally, the results were compared to determine the most suitable operating conditions in terms of yield, selectivity, residence time and reactor volume.

2.2.2. Reaction Network and Kinetic Model

The conversion of sorbitol to isosorbide takes place via a complex network of reactions comprising both sequential and parallel reactions. In the reaction mechanism, sorbitol is first converted into the 1,4-sorbitan intermediate, and isosorbide is then formed as a result of a

second dehydration step. However, undesirable side reactions also occur within the reaction system, causing various by-products to form. First two reactions represent desired pathway for isosorbide formation. Other reactions represent undesired pathways that decrease product yield and selectivity here.

In this study, the reaction network provided in the project guideline was used. The reactions are expressed as follows:



In terms of the objectives of this project, it has been assumed that all reactions are irreversible and first-order with respect to the reactant concentration. Accordingly, the reaction rate equations have been defined as follows:

$$r_1 = k_1 * C_{\text{sorbitol}}$$

$$r_2 = k_2 * C_{1,4-\text{sorbitan}}$$

$$r_3 = k_3 * C_{\text{sorbitol}}$$

$$r_4 = k_4 * C_{1,4-\text{sorbitan}}$$

$$r_5 = k_5 * C_{\text{isosorbide}}$$

Here, k_1 , k_2 , k_3 , k_4 and k_5 represent the rate constants of the related reactions; C_{sorbitol} , $C_{1,4-\text{sorbitan}}$ and $C_{\text{isosorbide}}$ represent the concentrations of sorbitol, 1,4-sorbitan and isosorbide, respectively.

2.2.3. Mole Balance Equations

In this study, a Plug Flow Reactor (PFR) has been selected as the reactor type. Under steady-state conditions, the overall mole balance for a PFR is expressed as follows:

$$In - Out + Generation - Consumption = Accumulation$$

If we consider a very small volume element ΔV within the reactor, the equation for this element is as follows:

$$F_{j|V} - F_{j|V+\Delta V} + r_j * \Delta V = 0$$

When we divide the equation by ΔV and take the limit as $\Delta V \rightarrow 0$, we obtain the PFR molar differential equation:

$$\frac{dF_j}{dV} = r_j$$

Here, F_j indicates the molar flow rate of component j , r_j indicates the reaction rate of component j , and V indicates the reactor volume.

Taking the reaction network into consideration, the stoichiometric equations for each species are written as follows. For clarity, the species definitions used in model are S = sorbitol, S1,4 = 1,4-sorbitan intermediate ISO = isosorbide and S1,5, O1 and O2 = undesired products.

Sorbitol (S)

Sorbitol is consumed in both the r_1 and r_3 reactions.

$$\frac{dF_S}{dV} = -r_1 - r_3$$

1,4-Sorbitan (S1,4)

1,4-Sorbitan is formed via the r_1 reaction and is consumed via the r_2 and r_4 reactions.

$$\frac{dF_{S1,4}}{dV} = r_1 - r_2 - r_4$$

Isosorbide (ISO)

It is formed by the r_2 reaction and is consumed by the r_5 reaction.

$$\frac{dF_{ISO}}{dV} = r_2 - r_5$$

1,5-Sorbitan (S_{1,5})

$$\frac{dF_{S_{1,5}}}{dV} = r_3$$

Others (O)

$$\frac{dF_O}{dV} = r_4 + r_5$$

2.2.4. Energy Balance

The conversion of sorbitol to isosorbide involves dehydration reactions and is described in the literature as an endothermic process. Consequently, heat must be supplied to the system for the reaction to proceed.

Generally, the energy balance of a PFR consists of the heat of reaction, the sensible heat of the fluid, and the heat transferred to the surroundings. For a reactor operating in a steady state, the overall energy balance can be expressed as follows:

$$\text{Energy input} - \text{Energy output} + \text{Heat released/consumed in the reaction} + \text{Heat transfer} = 0$$

However, in this project, the kinetic constants are given for specific temperatures, and the primary aim of the study is to investigate the effect of temperature on reactor performance. For this reason, each analysis was carried out for a specific constant temperature, and it was assumed that the reactor was isothermal.

Based on the assumption of isothermal operation, it was assumed that there was no temperature change along the reactor, and the temperature term was not included in the energy equation system. Accordingly, the temperature parameter was evaluated separately at values of 393.15 K, 403.15 K and 423.15 K.

For this reason, heat management has been addressed in detail in this preliminary design. As temperature fluctuations directly affect the kinetic constants and, consequently, the isosorbide yield, it is recommended that a jacketed tube reactor or an external heating system be used to maintain isothermal conditions.

2.2.5. Numerical Methodology

In this study, the species equations defining the conversion of sorbitol to isosorbide were derived on the basis of first-order reaction kinetics, and the changes in species concentrations along the reactor were determined by solving the resulting system of differential equations using numerical methods.

Firstly, the reaction rate equations and stoichiometric relationships were determined; next, the system equations were formulated using the given kinetic constants. In order to investigate the temperature-dependent variation in reactor performance, the analyses were carried out at temperatures of 393.15 K, 403.15 K and 423.15 K.

The analyses resulted in calculations of sorbitol conversion, intermediate product distribution and isosorbide formation; the data obtained were used to compare yields at different temperatures and to determine the optimum operating temperature. The system of differential equations was solved in MATLAB, and the results were presented in graphs and tables. Additionally, the computational approach and key design results were summarized in an Excel file.

3. THEORY

This section presents theoretical equations used in reactor model. Numerical substitutions and calculation results are given later in Design and Calculation Section.

3.1. Arrhenius Equation

The temperature dependence of the reaction rate constants is described by the Arrhenius equation.

$$k = A * e^{\frac{-E_a}{R*T}} \quad (1)$$

where k is the reaction rate constant, A is the pre-exponential factor in h⁻¹, E_a is the activation energy in j/mol, R is the universal gas constant in j/mol*K and T is the absolute temperature in K.

In this project, Arrhenius equation was used only to explain temperature dependence of rate constants. Kinetic constants were not calculated from the activation energy values; instead, rate constants provided in the project assignment were directly used for 393.15 K, 403.15 K and 423.15 K.

3.2. Reaction Network

Reaction network used in model consists of consecutive and parallel reactions. Species are defined as S for sorbitol, S1,4 for 1,4-sorbitan intermediate, ISO for isosorbide, and S1,5, O1 and O2 for undesired products.



First two reactions represent desired pathway for isosorbide production, and whereas the remaining reactions represent undesired side product formation.

3.3. PFR Species Balance

Overall species balance for PFR:

$$\frac{dC_i}{d\tau} = r_i \quad (7)$$

Or, more generally:

$$In - Out + Generation - Consumption = Accumulation$$

For a plug flow reactor, the molar balance for component j is:

$$\frac{dF_j}{dV} = r_j \quad (8)$$

For a liquid phase reactor with constant volumetric flow rate, balance can be written in terms of concentration and the residence time:

$$\tau = Q * V \quad (9)$$

where τ is the residence time, Q is the volumetric flow rate, V is the volume.

The concentration profiles inside the plug flow reactor were obtained using species balance equations derived from the reaction network.

$$\frac{dC_S}{d\tau} = -r_1 - r_3 \quad (10)$$

$$\frac{dC_{S1,4}}{d\tau} = r_1 - r_2 - r_4 \quad (11)$$

$$\frac{dC_{ISO}}{d\tau} = r_2 - r_5 \quad (12)$$

$$\frac{dC_{S1,5}}{d\tau} = r_3 \quad (13)$$

$$\frac{dC_{O1}}{d\tau} = r_4 \quad (14)$$

$$\frac{dC_{O2}}{d\tau} = r_5 \quad (15)$$

The inlet conditions used in the model were:

$$C_S(0) = 1 \frac{mol}{L} \quad (16)$$

$$C_{S1,4}(0) = C_{ISO}(0) = C_{S1,5}(0) = C_{O1}(0) = C_{O2}(0) = 0 \quad (17)$$

3.4. Reaction Rate Expressions

The reaction rates are assumed to follow first-order kinetics with respect to the reactant concentration.

$$r_1 = k_1 * C_S \quad (18)$$

$$r_2 = k_2 * C_{S1,4} \quad (19)$$

$$r_3 = k_3 * C_S \quad (20)$$

$$r_4 = k_4 * C_{S1,4} \quad (21)$$

$$r_5 = k_5 * C_{ISO} \quad (22)$$

where r_i represents the reaction rate of reaction i in mol/L*h, k_i represents the rate constants of reaction i in h^{-1} and C_i represents the concentration of the corresponding species in mol/L.

3.4. Euler Method

The system of ordinary differential equations was solved numerically using the Euler method.

$$C_{i,n+1} = C_{i,n} + \left(\frac{dC_i}{d\tau} \right)_n * \Delta\tau \quad (23)$$

where $C_{i,n+1}$ is the concentration of species i at next step, $C_{i,n}$ is the concentration of species i at current step, $\frac{dC_i}{d\tau}$ is the rate of concentration change with residence time and $\Delta\tau$ is the residence time step size used in the numerical solution.

In MATLAB calculation, residence-time step size was selected as:

$$\Delta\tau = 0.001h \quad (24)$$

3.5. Conversion

Sorbitol conversion was calculated using the following equation.

$$X_S = \frac{C_{S0} - C_S}{C_{S0}} \quad (25)$$

where X_S is the sorbitol conversion, C_{S0} is the initial sorbitol concentration in mol/L and C_S is the sorbitol concentration in mol/L at a given residence time.

3.6. Yield

Isosorbide yield was calculated as the amount of isosorbide formed relative to the initial amount of sorbitol:

$$Y_{ISO} = \frac{C_{ISO}}{C_{S0}} \quad (26)$$

where Y_{ISO} is the isosorbide yield, C_{ISO} is the isosorbide concentration in mol/L and C_{S0} is the initial sorbitol concentration in mol/L.

For reactor sizing, same yield definition was used to determine the required inlet sorbitol feed rate:

$$F_{S0} = \frac{F_{ISO}}{Y_{ISO}} \quad (27)$$

where F_{S0} is the inlet sorbitol molar flow rate (mol h⁻¹), F_{ISO} is the target isosorbide molar flow rate (mol h⁻¹), and Y_{ISO} is the isosorbide yield.

3.8. Selectivity

In this report, tabulated selectivity value is expressed as a desired to undesired product ratio rather than as percentage selectivity:

$$R_{D/U} = \frac{C_{ISO}}{C_{S1,5} + C_{O1} + C_{O2}} \quad (28)$$

where $R_{D/U}$ is the desired-to-undesired product ratio, C_{ISO} is the concentration of isosorbide in mol/L, $C_{S1,5}$ is the concentration of 1,5-sorbitan in mol/L, and C_{O1} and C_{O2} are the concentrations of the undesired by-products in mol/L.

A higher $R_{D/U}$ value indicates that more isosorbide is obtained relative to the total amount of undesired products.

Classic selectivity to converted sorbitol can also be expressed as:

$$S_{ISO} = \frac{Y_{ISO}}{X_S} \quad (29)$$

where S_{ISO} is the selectivity of isosorbide with respect to sorbitol conversion, Y_{ISO} is the isosorbide yield and X_S is the sorbitol conversion.

3.9. Reactor Volume Relation

After required inlet sorbitol flow rate was calculated, inlet volumetric flow rate was obtained from:

$$Q = \frac{F_{S0}}{C_{S0}} \quad (30)$$

where Q is the volumetric flow rate (m³*h⁻¹), F_{S0} is the inlet sorbitol molar flow rate (mol*h⁻¹), and C_{S0} is the inlet sorbitol concentration (mol*m⁻³).

The reactor volume was then calculated using the residence time:

$$V = Q * \tau \quad (31)$$

where V is the reactor volume in m³, Q is the volumetric flow rate in m³*h⁻¹ and τ is the residence time in h.

For preliminary reactor dimensioning, the reactor was assumed to be cylindrical:

$$V = \frac{\pi D^2}{4} L \quad (32)$$

where V is the reactor volume in m³, D is the reactor diameter in m, and L is the reactor length in m.

By using assumed length-to-diameter ratio:

$$\frac{L}{D} = 10 \quad (33)$$

where L is the reactor length in m and D is the reactor diameter in m.

Reactor diameter and length can be calculated as:

$$D = \left(\frac{4V}{10\pi} \right)^{1/3} \quad (34)$$

where D is the reactor diameter (m), and V is the reactor volume (m³).

3.10. Energy Balance / Isothermal Operation

For a non-isothermal PFR, energy balance generally includes heat of reaction, sensible heat change and heat transfer through the reactor wall. But, in this preliminary design, reactor was assumed to operate isothermally at each selected temperature.

$$\frac{dT}{d\tau} = 0 \quad (35)$$

This assumption means that sufficient heat is supplied to reactor to maintain constant operating temperature.

4. DESIGN AND CALCULATION

4.1. Parametric Study

In order to evaluate the effect of temperature on reactor performance, analyses were carried out at temperatures of 393.15 K, 403.15 K and 423.15 K. For each temperature, the maximum isosorbide formation, sorbitol conversion, selectivity, yield, residence time and by-product formation were calculated, and the corresponding results are presented in Table 1.

All performance values were evaluated at residence time where the isosorbide concentration reached its maximum for corresponding temperature. This approach was selected since isosorbide can also be consumed through undesired pathway represented by k_5 in longer residence times.

Table 1. Effect of Temperature on Reactor Performance

Temperature (K)	Conversion	Maximum Isosorbide (mol/L)	Yield (%)	Selectivity	Residence Time (h)	Total Undesired Products (mol/L)
393.15	1.0000	0.7398	73.98	3.370	0.224	0.2195
403.15	0.9988	0.7232	72.32	3.392	0.083	0.2132
423.15	0.9998	0.7492	74.92	3.448	0.033	0.2173

4.1.1. Effect of Temperature on Sorbitol Conversion

In order to investigate the effect of temperature on sorbitol conversion, conversion values were calculated at the point where isosorbide formation was highest for each temperature. The results indicate that the sorbitol conversion is very high at all temperatures. The conversion values obtained at 393.15 K, 403.15 K and 423.15 K were found to be 1.0000, 0.9988 and 0.9998, respectively.

Analysis of the results indicates that temperature has no significant effect on the conversion. This is because, at all temperatures examined, the reaction rates are high enough to consume the sorbitol almost entirely.

4.1.2. Effect of Temperature on Isosorbide Yield

Effect of temperature on isosorbide yield was evaluated using the maximum isosorbide concentrations. The maximum isosorbide concentrations obtained were calculated as 0.7398

mol/L, 0.7232 mol/L and 0.7492 mol/L at temperatures of 393.15 K, 403.15 K and 423.15 K, respectively. In contrast, the yields were found to be 73.98%, 72.32% and 74.92%, respectively.

The results show that the highest yield of isosorbide was obtained at 423.15 K among the temperatures examined. An increase in temperature led to a faster conversion of sorbitol into intermediates and subsequently into isosorbide by increasing the reaction rate constants. Consequently, the highest product formation occurred at 423.15 K.

4.1.3. Effect of Temperature on Product Selectivity

In order to examine the effect of temperature on product selectivity, selectivity values were calculated at the maximum isosorbide formation point for each temperature. The selectivity values obtained were found to be 3.370, 3.392 and 3.448 at temperatures of 393.15 K, 403.15 K and 423.15 K, respectively.

The results show that the selectivity of isosorbide increases as the temperature rises. The highest selectivity value was obtained at 423.15 K. This demonstrates that the desired reaction pathway becomes more predominant at high temperatures.

By-product formation was also evaluated. At the point where maximum isosorbide formation occurred, the total by-product concentrations were calculated as 0.2195 mol/L, 0.2132 mol/L and 0.2173 mol/L at temperatures of 393.15 K, 403.15 K and 423.15 K, respectively. Although the lowest total by-product formation was observed at 403.15 K, the isosorbide yield and selectivity obtained at this temperature were lower than the values at 423.15 K.

4.1.4. Effect of Temperature on Residence Time

Effect of temperature on the residence time required for maximum isosorbide formation was investigated. The analyses demonstrated that the residence times required to reach maximum isosorbide concentration at temperatures of 393.15 K, 403.15 K and 423.15 K were 0.224 hours, 0.083 hours and 0.033 hours, respectively.

The results show that the residence time required for maximum product formation decreases significantly as the temperature rises. The main reason for this is the increase in reaction rate constants at high temperatures, which results in the reactions proceeding more rapidly. In particular, at a temperature of 423.15 K, the system reached maximum isosorbide production in the shortest time.

4.2. Sample Calculation

This section presents the first Euler iteration of the calculations carried out, for example, at a temperature of 423.15 K.

Initial Conditions

The initial conditions are as follows:

$$C_{S0} = 1 \text{ mol/L}$$

$$C_{S1,4}(0) = C_{ISO}(0) = C_{S1,5}(0) = C_{O1}(0) = C_{O2}(0) = 0$$

Euler Step Size:

$$\Delta\tau = 0.001 \text{ h}$$

Kinetic constants at a temperature of 423.15 K:

$$k_1 = 221.90 \text{ h}^{-1}$$

$$k_2 = 103.31 \text{ h}^{-1}$$

$$k_3 = 8.80 \text{ h}^{-1}$$

$$k_4 = 12.12 \text{ h}^{-1}$$

$$k_5 = 4.94 \text{ h}^{-1}$$

Reaction Rates:

$$r_1 = k_1 * C_S$$

$$r_1 = 221.9 \text{ h}^{-1} * 1 \frac{\text{mol}}{\text{L}} = 221.9 \frac{\text{mol}}{\text{L} * \text{h}}$$

$$r_2 = k_2 * C_{S1,4}$$

$$r_2 = 103.31 \text{ h}^{-1} * 0 \frac{\text{mol}}{\text{L}} = 0 \frac{\text{mol}}{\text{L} * \text{h}}$$

$$r_3 = k_3 * C_S$$

$$r_3 = 8.8 \text{ h}^{-1} * 1 \frac{\text{mol}}{\text{L}} = 8.8 \frac{\text{mol}}{\text{L} * \text{h}}$$

$$r_4 = k_4 * C_{S1,4}$$

$$r_4 = 12.12 \text{ h}^{-1} * 0 \frac{\text{mol}}{\text{L}} = 0 \frac{\text{mol}}{\text{L} * \text{h}}$$

$$r_5 = k_5 * C_{ISO}$$

$$r_5 = 4.94 \text{ h}^{-1} * 0 \frac{\text{mol}}{\text{L}} = 0 \frac{\text{mol}}{\text{L} * \text{h}}$$

Species Derivatives:

The derivatives have been calculated using the species equations.

$$\frac{dC_S}{d\tau} = -r_1 - r_3 = -221.9 \frac{\text{mol}}{\text{L} * \text{h}} - 8.8 \frac{\text{mol}}{\text{L} * \text{h}} = -230.7 \frac{\text{mol}}{\text{L} * \text{h}}$$

$$\frac{dC_{S1,4}}{d\tau} = r_1 - r_2 - r_4 = 221.9 \frac{\text{mol}}{\text{L} * \text{h}} - 0 \frac{\text{mol}}{\text{L} * \text{h}} - 0 \frac{\text{mol}}{\text{L} * \text{h}} = 221.9 \frac{\text{mol}}{\text{L} * \text{h}}$$

$$\frac{dC_{ISO}}{d\tau} = r_2 - r_5 = 0 \frac{\text{mol}}{\text{L} * \text{h}} - 0 \frac{\text{mol}}{\text{L} * \text{h}} = 0 \frac{\text{mol}}{\text{L} * \text{h}}$$

$$\frac{dC_{S1,5}}{d\tau} = r_3 = 8.8 \frac{\text{mol}}{\text{L} * \text{h}}$$

$$\frac{dC_{O1}}{d\tau} = r_4 = 0 \frac{\text{mol}}{\text{L} * \text{h}}$$

$$\frac{dC_{O2}}{d\tau} = r_5 = 0 \frac{\text{mol}}{\text{L} * \text{h}}$$

Euler Method

The Euler method was applied using the following equation:

$$C_{new} = C_{old} + \frac{dC}{d\tau} * \Delta\tau$$

For Sorbitol;

$$C_{S-1} = 1 \frac{\text{mol}}{\text{L}} + \left(-230.7 \frac{\text{mol}}{\text{L} * \text{h}} \right) * 0.001 \text{ h} = 0.7693 \frac{\text{mol}}{\text{L}}$$

For 1,4-sorbitan;

$$C_{S1,4-1} = 0 \frac{\text{mol}}{\text{L}} + \left(221.9 \frac{\text{mol}}{\text{L} * \text{h}} \right) * 0.001 \text{ h} = 0.2219 \frac{\text{mol}}{\text{L}}$$

For Isosorbide;

$$C_{ISO-1} = 0 \frac{\text{mol}}{\text{L}} + \left(0 \frac{\text{mol}}{\text{L} * \text{h}} \right) * 0.001 \text{ h} = 0 \frac{\text{mol}}{\text{L}}$$

For 1,5-sorbitan;

$$C_{S1,5-1} = 0 \frac{\text{mol}}{\text{L}} + \left(8.8 \frac{\text{mol}}{\text{L} * \text{h}} \right) * 0.001 \text{ h} = 0.0088 \frac{\text{mol}}{\text{L}}$$

For other 1;

$$C_{O1-1} = 0 \frac{\text{mol}}{\text{L}} + \left(0 \frac{\text{mol}}{\text{L} * \text{h}} \right) * 0.001 \text{ h} = 0 \frac{\text{mol}}{\text{L}}$$

For other 2;

$$C_{O2-1} = 0 \frac{\text{mol}}{\text{L}} + \left(0 \frac{\text{mol}}{\text{L} * \text{h}} \right) * 0.001 \text{ h} = 0 \frac{\text{mol}}{\text{L}}$$

These calculations were repeated at all residence time steps until maximum isosorbide concentration was obtained.

The concentration profile obtained from MATLAB calculation for optimum temperature of 423.15 K is shown in the Figure 1.

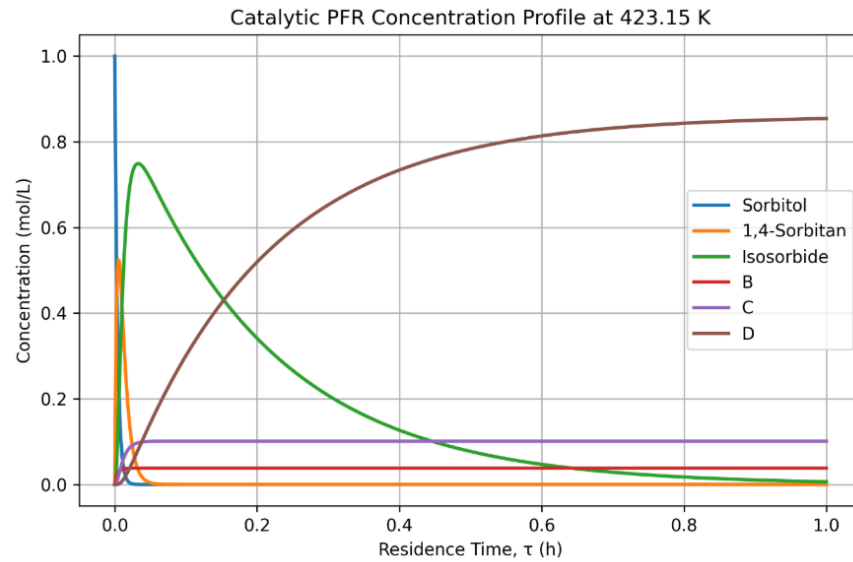


Figure 1. Concentration profiles of sorbitol, 1,4-sorbitan, isosorbide and undesired products in the catalytic PFR at 423.15 K.

For 423.15 K, maximum isosorbide concentration was found as:

$$C_{I,max} = 0.7492 \text{ mol/L}$$

at:

$$\tau_{opt} = 0.033 \text{ h}$$

Sorbitol conversion at this point is:

$$X_S = 0.99983$$

Isosorbide yield is:

$$Y_{ISO} = \frac{C_I}{C_{S0}} = \frac{0.7492}{1} = 0.7492$$

$$Y_{ISO} = 74.92\%$$

Total undesired product concentration is:

$$C_{undesired} = C_B + C_C + C_D = 0.2174 \text{ mol/L}$$

Desired-to-undesired ratio is:

$$R_{D/U} = \frac{C_I}{C_{undesired}} = \frac{0.7492}{0.2174} = 3.4468$$

Yearly production target is:

$$5000 \text{ ton/year}$$

By using 330 operating days per year:

$$t_{op} = 330 \times 24 = 7920 \text{ h/year}$$

The molecular weight of isosorbide here is:

$$MW_{ISO} = 146.14 \text{ g/mol}$$

The target isosorbide molar flow rate there is:

$$F_{ISO} = \frac{5000 \times 10^6}{146.14 \times 7920} = 4319.92 \text{ mol/h}$$

Since the yield is less than 100%, required inlet sorbitol molar flow rate is:

$$F_{S0} = \frac{F_{ISO}}{Y_{ISO}} = \frac{4319.92}{0.7492} = 5766.15 \text{ mol/h}$$

Since:

$$C_{S0} = 1 \text{ mol/L}$$

required volumetric flow rate is:

$$v_0 = \frac{F_{S0}}{C_{S0}} = 5766.15 \text{ L/h} = 5.766 \text{ m}^3/\text{h}$$

required reactor volume is:

$$V = v_0 \tau = (5.766)(0.033) = 0.1903 \text{ m}^3$$

For the preliminary dimensioning, $L/D = 10$ was assumed. Therefore

$$V = \frac{\pi D^2}{4} L$$

$$L = 10D$$

$$V = \frac{10\pi D^3}{4}$$

$$D = \left(\frac{4V}{10\pi} \right)^{1/3}$$

$$D = \left(\frac{4(0.1903)}{10\pi} \right)^{1/3} = 0.2894 \text{ m}$$

$$L = 10D = 2.8936 \text{ m}$$

Therefore, for optimum temperature of 423.15 K, the preliminary catalytic PFR dimensions are:

$$V = 0.1903 \text{ m}^3$$

$$D = 0.2894 \text{ m}$$

$$L = 2.8936 \text{ m}$$

4.3. Optimization

Optimization was carried out for 3 assigned temperature values. The primary aim of the optimization is to minimise the required reactor volume and the associated capital costs while achieving high isosorbide yield and selectivity.

The performance indicators obtained from the parametric study are presented in Table 2.

Table 2. Optimization Parameters at Different Temperatures

Temperature (K)	Yield (%)	Selectivity	Residence Time (h)	Relative Reactor Volume (L)	Reactor Volume (m ³)
393.15	73.98	3.370	0.224	6.8741	1.3080
403.15	72.32	3.392	0.083	2.6055	0.4958
423.15	74.92	3.448	0.033	1.0000	0.1903

Optimization results shows that increasing temperature significantly decreases required residence time. At 393.15 K, reactor requires 0.224 h to reach maximum isosorbide concentration. At 423.15 K, only 0.033 h is required. This causes significant and important decrease in reactor volume.

Effect of reaction temperature on isosorbide yield and selectivity to converted sorbitol is shown in Figure 2.

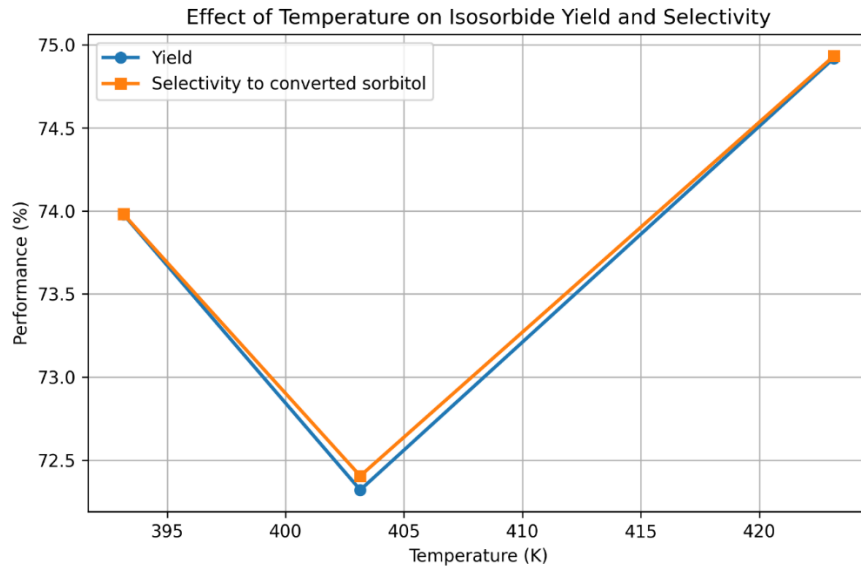


Figure 2. Effect of reaction temperature on isosorbide yield and selectivity to converted sorbitol

Reactor volume required at 393.15 K is approximately 6.87 times higher than reactor volume required at 423.15 K. Similarly, reactor volume required at 403.15 K is approximately 2.61 times higher than reactor volume required at 423.15 K.

Among three investigated temperatures, 423.15 K gives highest isosorbide yield, highest desired-to undesired ratio, shortest residence time, and lowest reactor volume. So therefore, 423.15 K was selected as optimum operating temperature for the final reactor design.

Reactor volume requirement decreased significantly as reaction temperature increased, as shown in the Figure 3.

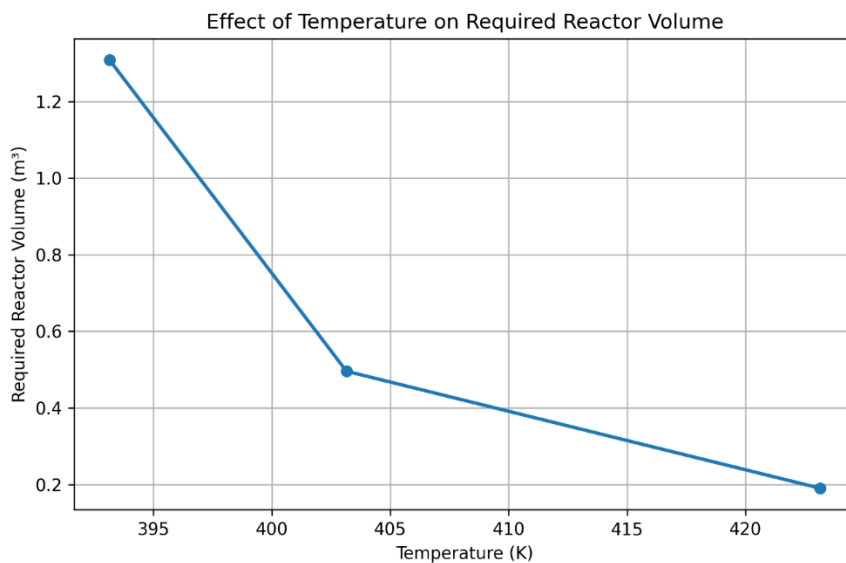


Figure 3. Effect of reaction temperature on required reactor volume

5. RESULTS AND DISCUSSION

Based on final calculation results, preliminary reactor design was summarized in the Table 4. This table presents the selected operating condition and corresponding reactor sizing values obtained from the temperature optimization study.

Table 3. Final Calculation Results

Design item	Selected/calculated value
Reactor type	Continuous liquid-phase catalytic PFR
Catalyst	H-Beta zeolite or MC-PhSO ₃ H solid acid catalyst
Operating temperature	423.15 K
Optimum residence time	0.033 h
Isosorbide yield	74.92%
Sorbitol conversion	99.98%
Desired-to-undesired ratio	3.4468
Target isosorbide flow rate	4319.92 mol/h
Required sorbitol feed rate	5766.15 mol/h
Volumetric flow rate	5.766 m ³ /h
Reactor volume	0.1903 m ³
Reactor diameter	0.2894 m
Reactor length	2.8936 m
Assumed L/D ratio	10

The final design was chosen at 423.15 K as it provided the highest isosorbide yield, the shortest residence time and the smallest reactor volume between the temperatures examined. Although the concentration of the undesirable product was not the lowest, the ratio of the desired product to the undesirable product was at its highest level; this made 423.15 K the most suitable operating temperature for the catalytic liquid-phase PFR.

Reactor volume was calculated by using yield corrected sorbitol feed rate instead of directly using the target isosorbide production rate. This gives more realistic preliminary design because part of the inlet sorbitol is consumed by side reactions. Thus, selected design represents best compromise between yield, selectivity, residence time, reactor sizing economy.

6. CONCLUSION

6. REFERENCES

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APPENDIX

```
clc;
clear;
close all;
%% CHE 302 Group 6 - Temperature Study for Catalytic PFR
% Irreversible first-order reaction network
% Explicit Euler integration with residence time tau
% Three assigned temperatures: 393.15 K, 403.15 K, 423.15 K
% The model calculates yield, selectivity, undesired products, feed rate,
% volumetric flow rate, reactor volume, diameter and length.
% Production basis: 5000 ton/year and 330 operating days/year.
% Catalyst effect is included in the supplied apparent first-order k-values.
%% Design basis
CS0 = 1.0;           % mol/L, initial sorbitol concentration
MW_ISO = 146.14;     % g/mol, molecular weight of isosorbide
production_ton_year = 5000; % ton/year isosorbide
operating_days = 330; % day/year
hours_per_day = 24; % h/day
operating_hours = operating_days * hours_per_day;
F_ISO_target = production_ton_year * 1e6 / MW_ISO / operating_hours; % mol/h
L_over_D = 10;       % assumed reactor length-to-diameter ratio
dtau = 0.001;        % h, Euler step size
tau_end = 1.0;       % h, simulation range
N = round(tau_end/dtau) + 1;
%% Temperature and kinetic data from project assignment
T_values = [393.15; 403.15; 423.15];
k_values = [ ...
    43.83, 14.29, 1.41, 1.67, 0.79; ...
    76.20, 38.05, 1.97, 2.60, 3.30; ...
    221.90, 103.31, 8.80, 12.12, 4.94];
%% Result storage
results = zeros(length(T_values), 15);
```

```

allProfiles = cell(length(T_values),1);
for t = 1:length(T_values)
    T = T_values(t);
    k1 = k_values(t,1);
    k2 = k_values(t,2);
    k3 = k_values(t,3);
    k4 = k_values(t,4);
    k5 = k_values(t,5);
    %% Allocate arrays
    tau = zeros(N,1);
    Cs = zeros(N,1); % Sorbitol
    Ca = zeros(N,1); % 1,4-sorbitan intermediate
    Ci = zeros(N,1); % Isosorbide
    Cb = zeros(N,1); % Side product branch from sorbitol
    Cc = zeros(N,1); % Others from intermediate
    Cd = zeros(N,1); % Others from isosorbide
    %% Initial conditions
    Cs(1) = CS0;
    Ca(1) = 0;
    Ci(1) = 0;
    Cb(1) = 0;
    Cc(1) = 0;
    Cd(1) = 0;
    %% Euler method
    for i = 1:N-1
        r1 = k1 * Cs(i);
        r2 = k2 * Ca(i);
        r3 = k3 * Cs(i);
        r4 = k4 * Ca(i);
        r5 = k5 * Ci(i);
        dCs = -(r1 + r3);
        dCa = r1 - r2 - r4;
        dCi = r2 - r5;
        dCb = r3;

```

```

dCc = r4;
dCd = r5;
Cs(i+1) = Cs(i) + dCs * dtau;
Ca(i+1) = Ca(i) + dCa * dtau;
Ci(i+1) = Ci(i) + dCi * dtau;
Cb(i+1) = Cb(i) + dCb * dtau;
Cc(i+1) = Cc(i) + dCc * dtau;
Cd(i+1) = Cd(i) + dCd * dtau;
tau(i+1) = tau(i) + dtau;
end
%%% Performance indicators at maximum isosorbide concentration
[maxCi, idx] = max(Ci);
tau_opt = tau(idx);
conversion = (CS0 - Cs(idx)) / CS0;
yield_iso = maxCi / CS0;           % fraction
selectivity_conv = yield_iso / conversion; % fraction to converted sorbitol
total_undesired = Cb(idx) + Cc(idx) + Cd(idx);
desired_to_undesired = maxCi / total_undesired;
%%% Feed and reactor sizing
% Since yield < 100%, inlet sorbitol molar flow must be higher than target isosorbide flow.
F_S0_required = F_ISO_target / yield_iso;    % mol/h
v0_L_h = F_S0_required / CS0;                % L/h
v0_m3_h = v0_L_h / 1000;                    % m^3/h
V_m3 = v0_m3_h * tau_opt;                   % m^3
D_m = (4 * V_m3 / (pi * L_over_D))^(1/3);    % m, from V = pi*D^2/4*L and L=10D
L_m = L_over_D * D_m;                       % m
results(t,:) = [T, maxCi, tau_opt, conversion, yield_iso*100, ...
    selectivity_conv*100, Cb(idx), Cc(idx), Cd(idx), total_undesired, ...
    desired_to_undesired, F_S0_required, v0_m3_h, V_m3, D_m];
allProfiles{t} = table(tau, Cs, Ca, Ci, Cb, Cc, Cd);
%%% Plot concentration profile for each temperature
figure('Name', sprintf('Concentration profile %.2f K', T));
plot(tau, Cs, 'LineWidth', 2); hold on;
plot(tau, Ca, 'LineWidth', 2);

```

```

plot(tau, Ci, 'LineWidth', 2);
plot(tau, Cb, 'LineWidth', 2);
plot(tau, Cc, 'LineWidth', 2);
plot(tau, Cd, 'LineWidth', 2);
xlabel('Residence Time, \tau (h)');
ylabel('Concentration (mol/L)');
title(sprintf('Catalytic PFR Concentration Profile at %.2f K', T));
legend('Sorbitol','1,4-Sorbitan','Isosorbide','B','C','D','Location','best');
grid on;
end
%%% Results table
ResultsTable = array2table(results, 'VariableNames', { ...
    'Temperature_K', 'Max_Isosorbide_mol_L', 'Residence_Time_h', ...
    'Conversion', 'Yield_percent', 'Selectivity_to_Converted_S_percent', ...
    'Cb_mol_L', 'Cc_mol_L', 'Cd_mol_L', 'Total_Undesired_mol_L', ...
    'Desired_to_Undesired_Ratio', 'Required_Sorbitol_Feed_mol_h', ...
    'Volumetric_Flow_m3_h', 'Reactor_Volume_m3', 'Diameter_m'});
ResultsTable.Length_m = L_over_D * ResultsTable.Diameter_m;
ResultsTable.Relative_Volume      =      ResultsTable.Reactor_Volume_m3      ./
min(ResultsTable.Reactor_Volume_m3);
[~, optimumIndex] = max(ResultsTable.Yield_percent);
OptimumCondition = ResultsTable(optimumIndex,:);
disp('Group 6 Temperature Study Results');
disp(ResultsTable);
disp('Optimum condition within the three assigned temperatures:');
disp(OptimumCondition);
%%% Comparative plot
figure('Name','Temperature optimization summary');
plot(ResultsTable.Temperature_K, ResultsTable.Yield_percent, '-o', 'LineWidth', 2); hold on;
plot(ResultsTable.Temperature_K, ResultsTable.Selectivity_to_Converted_S_percent, '-s',
'LineWidth', 2);
xlabel('Temperature (K)');
ylabel('Performance (%)');
title('Effect of Temperature on Isosorbide Yield and Selectivity');

```

```
legend('Yield','Selectivity to converted sorbitol','Location','best');  
grid on;  
%% Export table  
writetable(ResultsTable, 'group6_temperature_results.csv');
```